

Dietary correlates of urinary 6-sulfatoxymelatonin concentrations in the Nurses' Health Study cohorts

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Background: Age and certain lifestyle factors, including a higher body mass index and exposure to light at night, are related to lower circulating concentrations of melatonin—a hormone with probable cancer-protective properties. Although melatonin is a direct derivative of the essential amino acid tryptophan, little is known about the relation of diet with melatonin concentrations. Objective: The objective was to examine cross-sectional associations of various nutrients and dietary factors as well as food groups with creatinine-adjusted first morning urinary melatonin (6-sulfatoxymelatonin; aMT6s) concentrations. Design: Participants were 998 healthy women from 2 independent cohorts: the Nurses' Health Study (NHS; n = 585) and NHS II (n = 413). We computed least-squares mean hormone concentrations across categories of dietary variables, with adjustment for total energy intake, age, and other nondietary factors known to be associated with aMT6s concentrations. Results: In multivariate analyses, we found no significant associations between the intake of various nutrients, including tryptophan and urinary melatonin concentrations. A higher intake of meat, particularly red meat, was associated with lower concentrations of aMT6s (adjusted mean concentrations of aMT6s across increasing quartiles of red meat intake were 17.9, 17.0, 18.1, and 15.3 ng/mg creatinine; P for trend = 0.02). In contrast, neither poultry intake (including turkey) nor fish intake was associated with aMT6s concentrations. Conclusion: Although no specific nutrients were associated with altered concentrations of melatonin, our findings raise the possibility that several specific foods, including red meat, could affect cancer risk through the lowering of melatonin concentrations.